```python
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import classification_report
import re
import joblib
import string
```

✓ 0.0s

```python
fake = pd.read_csv('Fake.csv')
true = pd.read_csv('True.csv')
```

✓ 2.4s

```
fake.head()
```

✓ 0.0s

	title	text	subject	date
0	Donald Trump Sends Out Embarrassing New Year'...	Donald Trump just couldn t wish all Americans ...	News	December 31, 2017
1	Drunk Bragging Trump Staffer Started Russian ...	House Intelligence Committee Chairman Devin Nu...	News	December 31, 2017
2	Sheriff David Clarke Becomes An Internet Joke...	On Friday, it was revealed that former Milwauk...	News	December 30, 2017
3	Trump Is So Obsessed He Even Has Obama's Name...	On Christmas day, Donald Trump announced that ...	News	December 29, 2017
4	Pope Francis Just Called Out Donald Trump Dur...	Pope Francis used his annual Christmas Day mes...	News	December 25, 2017

```
true.head()
```

✓ 0.0s

	title	text	subject	date
0	As U.S. budget fight looms, Republicans flip t...	WASHINGTON (Reuters) - The head of a conservat...	politicsNews	December 31, 2017
1	U.S. military to accept transgender recruits o...	WASHINGTON (Reuters) - Transgender people will...	politicsNews	December 29, 2017
2	Senior U.S. Republican senator: 'Let Mr. Muell...	WASHINGTON (Reuters) - The special counsel inv...	politicsNews	December 31, 2017
3	FBI Russia probe helped by Australian diplomat...	WASHINGTON (Reuters) - Trump campaign adviser ...	politicsNews	December 30, 2017
4	Trump wants Postal Service to charge 'much mor...	SEATTLE/WASHINGTON (Reuters) - President Donal...	politicsNews	December 29, 2017

+ Code + Markdown

```
data = pd.concat([fake,true], axis = 0)
```

0.0s

```
data.sample(10)
```

0.0s

	title	text	subject	date	class
2569	HILLARY PLAYS VICTIM CARD...Here's Why That's NO...	The Clintons always blame somebody else. Now i...	politics	Oct 30, 2016	0
9202	JUST IN: Documents Reveal Hillary Clinton "Str...	Here s yet another reason to be very suspiciou...	politics	Dec 14, 2017	0
228	Trump Finalizes Betrayal Of LGBT Community Wi...	After a campaign in which he shamelessly pande...	News	October 12, 2017	0
3894	Polish MPs back judicial overhaul seen by EU a...	WARSAW (Reuters) - Poland s ruling party lawma...	worldnews	November 24, 2017	1
2021	Democrats Offer Bill That Would FORCE Trump T...	The vast majority of Donald Trump s campaign p...	News	March 24, 2017	0
4782	Trump mum on healthcare plan's future if it fa...	WASHINGTON (Reuters) - U.S. President Donald T...	politicsNews	March 22, 2017	1
2733	Iran's defense minister to visit Russia soon: RIA	MOSCOW (Reuters) - Iranian Defence Minister Am...	worldnews	December 8, 2017	1
5920	Grief-stricken families of Japanese abducted b...	TOKYO (Reuters) - Takuya Yokota vividly rememb...	worldnews	November 1, 2017	1
0592	GERMANY: 10,000 Muslims Allegedly Registered T...	After allegedly registering 10,000 Muslims to ...	politics	Jun 17, 2017	0
4255	U.S. Secretary of State Tillerson made $27.4 m...	HOUSTON (Reuters) - U.S. Secretary of State Re...	politicsNews	April 13, 2017	1

```
data.reset_index(inplace=True)
```

0.0s

```
data.drop(['index'],axis = 1,inplace=True)
```

0.0s

```
data.sample(5)
```

0.0s

	text	class
3635	WASHINGTON (Reuters) - Republican Senator John...	1
5122	FRANKFURT (Reuters) - A suspicious package tha...	1
7113	WASHINGTON (Reuters) - U.S. President Donald T...	1
4142	This is a MUST WATCH!According to American New...	0
3536	ANKARA (Reuters) - France s foreign minister c...	1

```
def clean_text(text):
    text = text.lower()
    text = re.sub('\[.*?\]',"",text)
    text = re.sub("\\W"," ",text)
    text = re.sub("https?://\S+|www\.\S+","",text)
    text = re.sub("<.*?>+","",text)
    text = re.sub("[%s]" % re.escape(string.punctuation),"",text)
    text = re.sub("\n","",text)
    text = re.sub("\w*\d\w*","",text)
```

```
data["text"] = data["text"].apply(clean_text)
```

✓ 40.6s

```
x=data["text"]
y=data["class"]

xtrain, xtest, ytrain, ytest = train_test_split(x,y,test_size=0.25,random_state=42)
```

✓ 0.0s

```
vectorizer = TfidfVectorizer()
xv_train = vectorizer.fit_transform(xtrain)
xv_test = vectorizer.transform(xtest)
```

✓ 27.6s

```
xv_train = vectorizer.fit_transform(xtrain)
xv_test = vectorizer.transform(xtest)
```

✓ 27.6s

```
lr = LogisticRegression()
lr.fit(xv_train,ytrain)
```

✓ 9.0s

```
LogisticRegression()
```

```
prediction = lr.predict(xv_test)
lr.score(xv_test,ytest)
```

✓ 0.0s

```
0.9860133630289533
```

```
prediction = lr.predict(xv_test)
lr.score(xv_test,ytest)
```

0.0s

```
9860133630289533
```

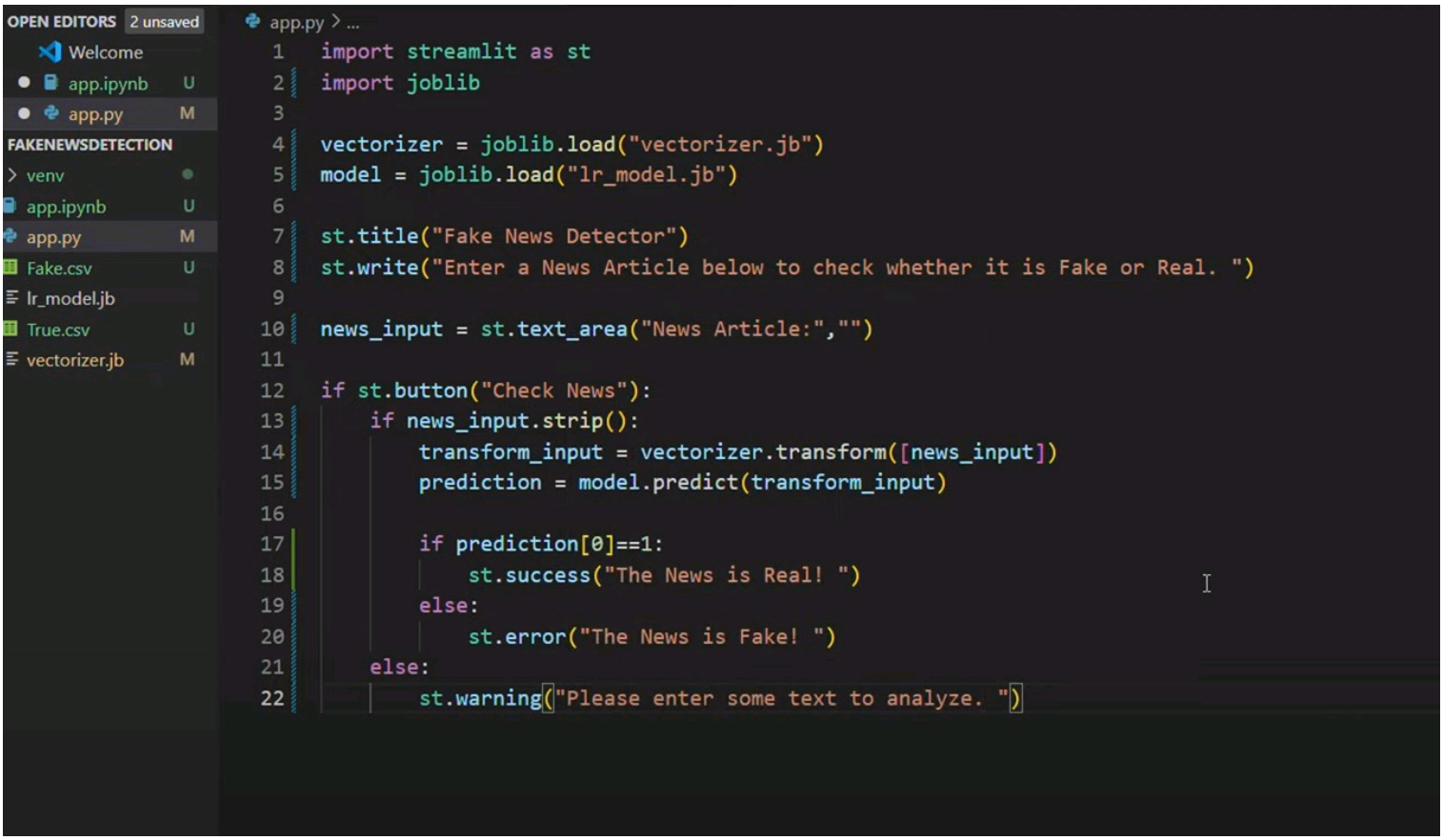
```
print(classification_report(ytest,prediction))
```

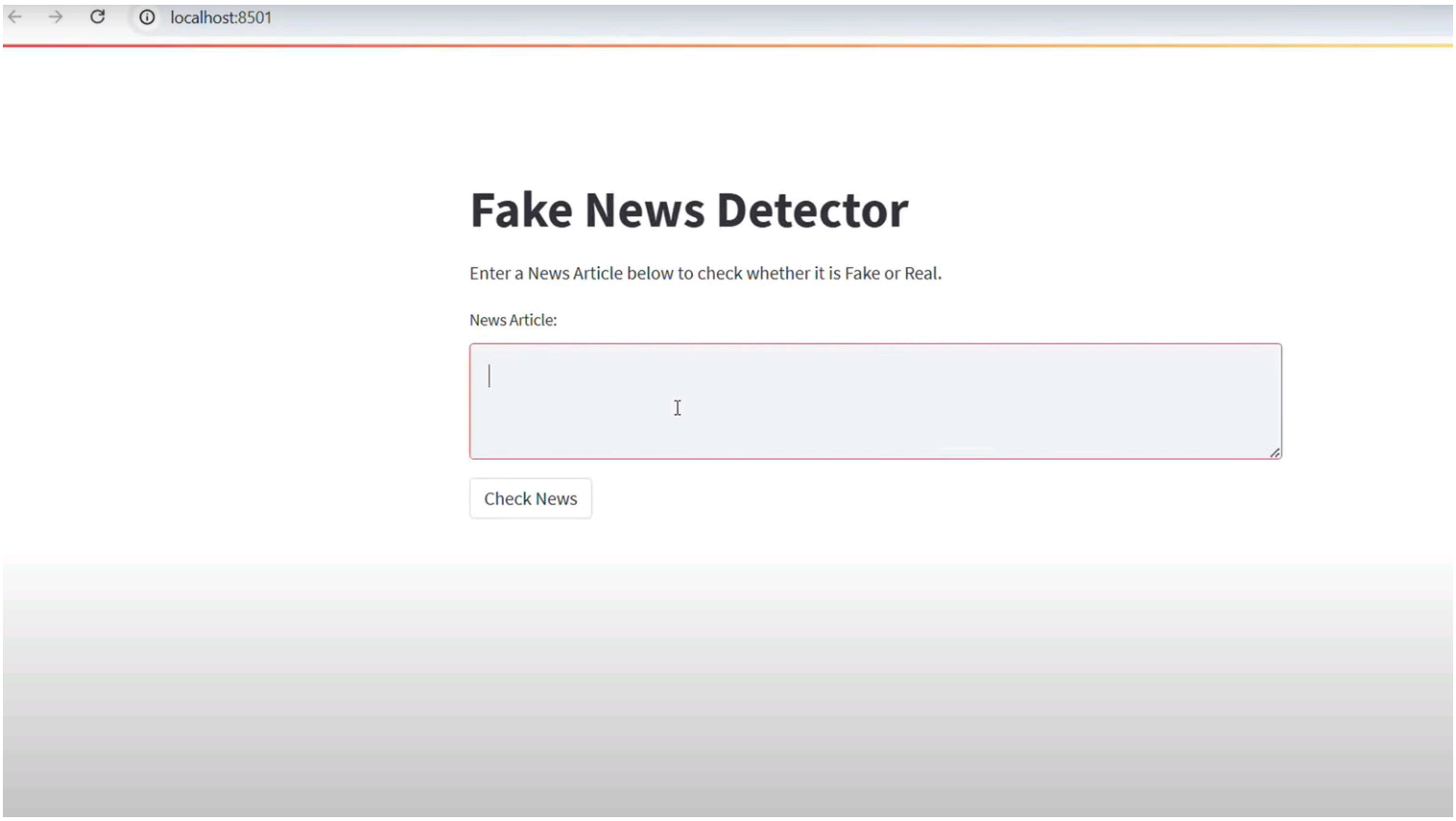
0.1s

```
              precision    recall  f1-score   support

           0       0.99      0.99      0.99      5895
           1       0.98      0.99      0.99      5330

    accuracy                           0.99     11225
   macro avg       0.99      0.99      0.99     11225
ighted avg       0.99      0.99      0.99     11225
```


OPEN EDITORS 2 unsaved
Welcome
app.ipynb U
app.py M
FAKENEWSDETECTION
venv
app.ipynb U
app.py M
Fake.csv U
lr_model.jb
True.csv U
vectorizer.jb M
app.py > ...
1 import streamlit as st
2 import joblib
3
4 vectorizer = joblib.load("vectorizer.jb")
5 model = joblib.load("lr_model.jb")
6
7 st.title("Fake News Detector")
8 st.write("Enter a News Article below to check whether it is Fake or Real. ")
9
10 news_input = st.text_area("News Article:","")
11
12 if st.button("Check News"):
13 if news_input.strip():
14 transform_input = vectorizer.transform([news_input])
15 prediction = model.predict(transform_input)
16
17 if prediction[0]==1:
18 st.success("The News is Real! ")
19 else:
20 st.error("The News is Fake! ")
21 else:
22 st.warning("Please enter some text to analyze. ")


localhost:8501
Fake News Detector
Enter a News Article below to check whether it is Fake or Real.
News Article:
Check News

```
" Trump Said Some INSANELY Racist Stuff Inside The Oval Office, And Witnesses Back It Up","In the wake of yet another court dec
" Former CIA Director Slams Trump Over UN Bullying, Openly Suggests He's Acting Like A Dictator (TWEET)","Many people have rais
 WATCH: Brand-New Pro-Trump Ad Features So Much A** Kissing It Will Make You Sick,"Just when you might have thought we d get a
" Papa John's Founder Retires, Figures Out Racism Is Bad For Business","A centerpiece of Donald Trump s campaign, and now his p
 WATCH: Paul Ryan Just Told Us He Doesn't Care About Struggling Families Living In Blue States,"Republicans are working overtim
 Bad News For Trump — Mitch McConnell Says No To Repealing Obamacare In 2018,"Republicans have had seven years to come up with
" WATCH: Lindsey Graham Trashes Media For Portraying Trump As 'Kooky,' Forgets His Own Words","The media has been talking all d
 Heiress To Disney Empire Knows GOP Scammed Us - SHREDS Them For Tax Bill,"Abigail Disney is an heiress with brass ovaries who
 Tone Deaf Trump: Congrats Rep. Scalise On Losing Weight After You Almost Died,"Donald Trump just signed the GOP tax scam into
 The Internet Brutally Mocks Disney's New Trump Robot At Hall Of Presidents,"A new animatronic figure in the Hall of Presidents
 Mueller Spokesman Just F-cked Up Donald Trump's Christmas,"Trump supporters and the so-called president s favorite network are
 SNL Hilariously Mocks Accused Child Molester Roy Moore For Losing AL Senate Race (VIDEO),"Right now, the whole world is lookin
 Republican Senator Gets Dragged For Going After Robert Mueller,"Senate Majority Whip John Cornyn (R-TX) thought it would be a
" In A Heartless Rebuke To Victims, Trump Invites NRA To Xmas Party On Sandy Hook Anniversary","It almost seems like Donald Tru
 KY GOP State Rep. Commits Suicide Over Allegations He Molested A Teen Girl (DETAILS),"In this #METOO moment, many powerful men
 Meghan McCain Tweets The Most AMAZING Response To Doug Jones' Win In Deep-Red Alabama,"As a Democrat won a Senate seat in deep
 CNN CALLS IT: A Democrat Will Represent Alabama In The Senate For The First Time In 25 Years,"Alabama is a notoriously deep re
 White House: It Wasn't Sexist For Trump To Slut-Shame Sen. Kirsten Gillibrand (VIDEO),"A backlash ensued after Donald Trump la
 Despicable Trump Suggests Female Senator Would 'Do Anything' With Him For Campaign Money (TWEET),"Donald Trump is afraid of st
 Accused Child Molesting Senate Candidate Roy Moore Sides With Putin Over Reagan (VIDEO),"Ronald Reagan is largely seen as the
" WATCH: Fox Host Calls For A 'Cleansing' Of The FBI, And To Arrest Everyone Investigating Trump"," Judge  Jeanine Pirro has co
 Liberal Group Trolls Trump At Roy Moore Rally In The Best Possible Way (VIDEO),"Donald Trump held a rally for Alabama Senate c
" Don Jr. Tries To Mock Al Franken's Resignation, Backfires Immediately","When Sen. Al Franken (D-MN) announced his plans to re
```

OUTPUT DEBUG CONSOLE TERMINAL PORTS JUPYTER streamlit

Fake News Detector

Enter a News Article below to check whether it is Fake or Real.

News Article:

Senator Warren hits out at 'effort to politicize' U.S. consumer agency."WASHINGTON (Reuters) - Democratic Senator Elizabeth Warren is taking aim at budget chief Mick Mulvaney's plan to fill the ranks of the U.S. consumer financial watchdog with political allies, according to letters seen by Reuters, the latest salvo in a broader battle over who should run the bureau. President Donald

Check News

The News is Real!

Fake News Detector

Enter a News Article below to check whether it is Fake or Real.

News Article:

they are supposed to be. It's time to get out and flip the House and possibly the Senate in 2018, and resist in the meantime, if we want to save our country from devolving into something that looks more like Russia or North Korea than the America we have always know. We re already well on our way.Featured image via BRENDAN SMIALOWSKI/AFP/Getty Images",News,"December 22, 2017"

Check News

The News is Fake!